

Disaster Management System By Remotz Systems

Deliverable-2

Project Team

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Semester 6

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PART C - Deliverable 2:

1. Introduction

The Disaster Management System is a comprehensive software solution designed to streamline coordination between citizens, NGOs, rescue teams, and administrators during crisis situations. By digitizing the reporting and response process, the system aims to minimize response times and ensure resources are allocated efficiently to areas in need.

2. Project Vision

To create a centralized, resilient platform that empowers communities and response organizations to collaborate effectively during disasters, ultimately saving lives and reducing property damage through timely information and organized action.

3. Intended Use of the System

Stakeholders and Needs

- **Citizens (Main Beneficiaries):**
 - **Need:** To quickly report disasters, request urgent assistance, and receive real-time updates on safety status.
 - **Usage:** They will use the mobile/web interface to submit reports and track their requests.
- **NGOs & Rescue Teams:**
 - **Need:** To view a verified list of disasters, accept rescue missions, and provide feedback on the ground situation.
 - **Usage:** They will log in to view assigned tasks, update status, and coordinate with the admin.
- **Admin:**
 - **Need:** Centralized control to verify incidents, mobilize resources (assign NGOs/Rescue teams), and analyze post-disaster data.
 - **Usage:** They will use the dashboard to manage users, records, and generate reports.

4. Features / Overall Functionality

Completed in Sprint 1 (Foundation):

- Secure role-based User Authentication and Registration.
- Citizen-driven Disaster Reporting and Admin-driven Disaster Record Management

(CRUD).

- Real-time Alert Broadcasting to keep stakeholders informed.

New Features in Sprint 2 (Enhancements):

- **Centralized Resource Inventory:** Admins can now add, edit, and track master supplies (e.g., tents, medical kits, food).
- **Dynamic Resource Allocation:** Admins can assign specific quantities of resources directly to dispatched teams.
- **Field Resource Requests:** NGOs and Rescue Teams can view their current allocations and request further assistance dynamically from the field.
- **Emergency SOS:** A dedicated, high-priority emergency trigger for citizens in immediate danger.

How Sprint 2 Improves System Capability: Sprint 2 bridges the critical gap between *knowing* a disaster happened and responding to it. By digitizing resource logistics, the DMS prevents double-allocation of critical supplies, eliminates communication bottlenecks between the Admin and ground teams, and allows citizens to flag life-or-death emergencies separate from general disaster reports.

5. User Stories

The following user stories define the functional requirements for Sprint 2:

Sprint 2 - Resource Management & Emergency Response

- **US-11:** Manage Resources (Admin)
 - **US-12:** Add Resource (Admin)
 - **US-13:** Edit Resource (Admin)
 - **US-14:** Delete Resource (Admin)
 - **US-15:** Allocate Resource to NGO/Rescue Team (Admin)
- **US-16:** View Allocated Resources (NGO, Rescue Team)
- **US-17:** Request Additional Resources (NGO, Rescue Team)
- **US-18:** Call for Emergency (Citizen)

6. NFR Specification (Natural Language)

- **Performance:** The system shall send broadcast alerts to all stakeholders within 5 seconds of the Admin initiating the alert.
- **Availability:** The system must be available 99.9% of the time to ensure citizens can

report disasters at any moment.

- **Security:** Only users with "Admin" role privileges can access the "Manage Disaster Records" and "Broadcast Alert" modules; passwords must be encrypted in the database.
- **Usability:** The "Report Disaster" form should be navigable and submittable within 3 clicks to ensure ease of use during panic situations.

7. Iteration 1 Implementation Screenshots

7.1 Manage Resources

DMS - Manage Resources

Disaster Management System

Admin UserLogout

MENU

- Dashboard
- Send Alert
- Manage Disasters
- Assign NGO
- Assign Rescue Team
- Track Rescue Teams
- Manage Resources
- Generate Report
- Review Feedback

Manage Resources

Track inventory and allocate disaster relief resources

+ Add Resource

Inventory

Requested Resources

Filter by Type:

AllMedicalFoodShelterEquipment

Allocated: 15 Units

Medical Supplies Kit

Medical

Available

145 / 160 Units

Add StockDetails

Allocated: 28 Units

Emergency Tents

Shelter

Available

47 / 75 Units

Add StockDetails

Water Purification Units

Equipment

Available

20 / 20 Units

Add StockDetails

Allocated: 20 Units

First Aid Kits

Medical

Add StockDetails

Rescue Boats

Equipment

Add StockDetails

Allocated: 150 Boxes

Food Packages

Food

Add StockDetails

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Requested Resources

Filter by Status:

AllPendingAllocatedRejected

ID	Resource	Quantity	Requester	Priority	Status	Requested Date	Action
PKG001	Medical Supplies Kit (15 Un...	15 Units	Pakistan Red Cresc...	High	Allocated	Nov 22, 2025 05:5...	
PKG002	Emergency Tents (10 Units)	10 Units	Alpha Rescue Squad	Critical	Rejected	Nov 22, 2025 05:5...	
PKG003	First Aid Kits (20 Units)	20 Units	Red Crescent Karachi	Medium	Allocated	Nov 22, 2025 05:5...	
PKG004	Food Packages (50 Boxes)	50 Boxes	Beta Rescue Team	High	Allocated	Nov 22, 2025 05:5...	
PKG005	Blankets (50 Units)	50 Units	Pakistan Red Cresc...	High	Allocated	Nov 22, 2025 06:2...	
PKG006	Food Packages (100 Boxes)	100 Boxes	RT005	Critical	Allocated	Nov 22, 2025 07:4...	
PKG007	Emergency Tents (50 Units)	50 Units	RT006	Critical	Rejected	Nov 22, 2025 08:2...	

8. Work Division

- **Moiz Ansari:** Analyzed and figured out the required user stories. Divided tasks among members. Updated documents for sprint 2. Created a Trello board. Developed Manage Resources use case and some other files.
 - **Rehan Tajik:** Developed US-16 to US-18 and sub-stories (CRUD operations).
 - **Zohaib Umer:** Developed US-11 to US-15 and sub-stories (CRUD operations).
 - **Tayyab:** Wrote all structured specifications.
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